# MILITARY SPECIFICATION

## ANTENNA AT-983/ARN AND ANTENNA AT-983A/ARN

### 1. SCOPE

1.1 This specification covers the requirements for two configurations of airborne glide slope receiving antennas, designated Antenna AT-983/ARN and Antenna AT-983A/ARN.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-17 | Cables, Radiofrequency, Coaxial, Dual Coaxial, Twin Conductor and Twin Lead |
| MIL-E-5400 | Electronic Equipment, Airborne, General Specification for |
| MIL-C-39012 | Connector, Coaxial Radio Frequency, General Specification for |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-831 | Test Reports, Preparation of |

(Copies of documents required by contractor in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

### 3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

3.2 General specification. The requirements of MIL-E-5400 shall apply as requirements of this specification with additions or exceptions to the paragraphs of MIL-E-5400, as identified by similarity of paragraph headings, called out herein. When the two documents conflict, this specification shall govern.

3.2.1 Classification. The AT-983/ARN and AT-983A/ARN antennas shall be class 2 equipment as defined in MIL-E-5400.

3.2.2 Approval of nonstandard parts and materials. Contract requirements for nonstandard parts and materials approval shall be category III as defined in MIL-E-5400.

3.2.3 Electromagnetic interference characteristics. The requirements for electromagnetic interference shall not apply.

3.2.4 Maintainability. Maintainability requirements shall not apply.

3.2.5 Reliability. Reliability requirements shall not apply.

3.3 Design and construction. The Antenna AT-983/ARN shall be essentially designed and constructed in accordance with figure 1. The Antenna AT-983A/ARN shall be essentially designed and constructed in accordance with figure 2.

3.3.1 m Dimensions and weight. Dimensions of the antennas shall be in accordance with the applicable figure. The weight of the antennas shall not exceed 0.5 pound.

3.3.2 Transmission line and connector. The antennas shall use a UG-680/U receptacle, MIL-C-39012, and shall achieve specified performance when connected to the associated receiver by means of coaxial cable RG-58/U, MIL-C-17, or equal.

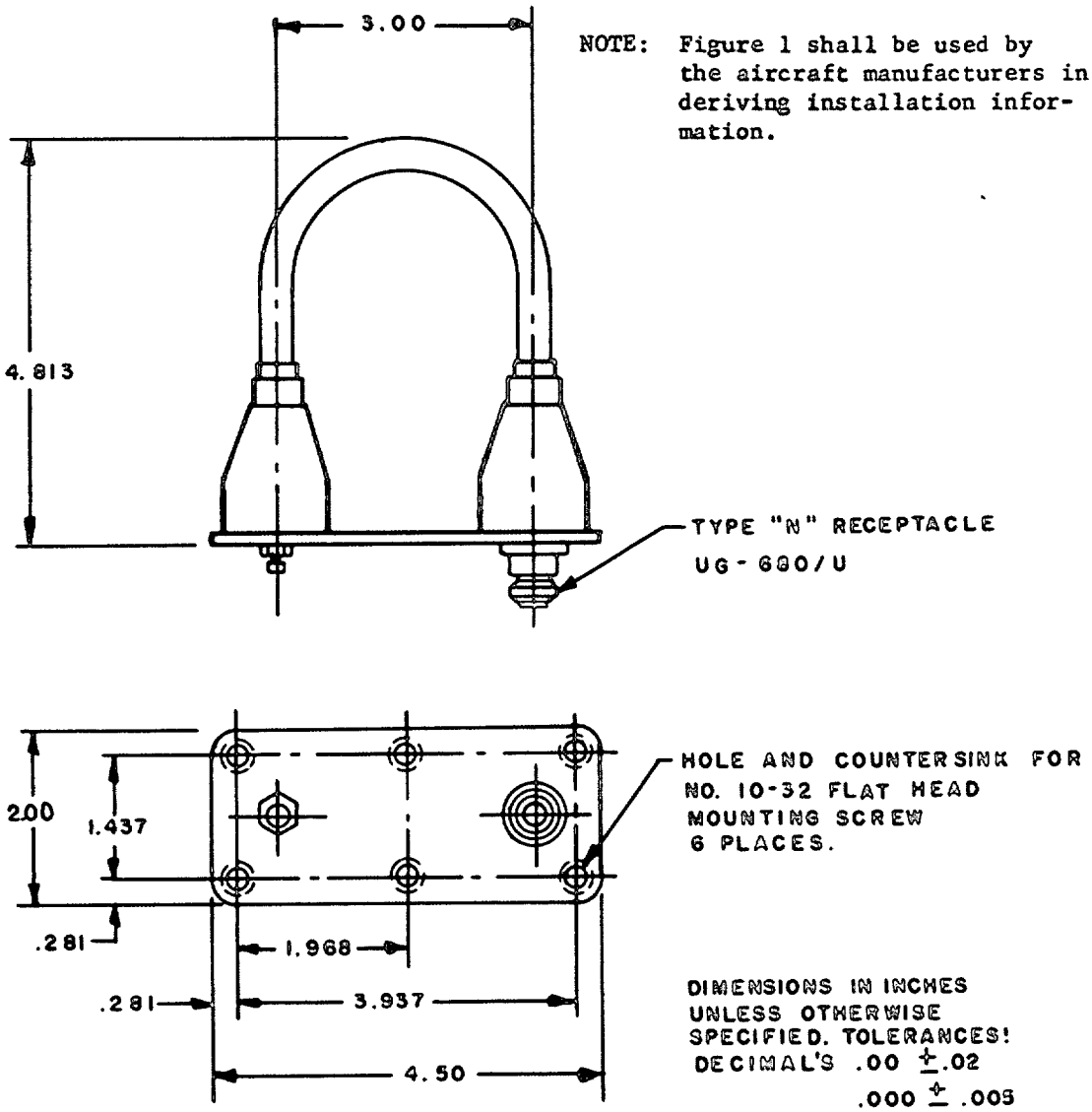
3.4 Performance. The detailed performance requirements specified herein shall be achieved under any probable combination of the environmental conditions specified for class 2 equipment.

3.4.1 Response. The antennas shall cover the frequency range from 328.6 to 335.4 MHz with a voltage standing wave ratio (VSWR) no greater than 3:1 under standard room conditions. During any combination of environmental conditions, the VSWR shall not exceed 5:1.

3.4.2 Polarization. The antennas shall achieve its required performance when receiving radio signals which are at least 95 percent horizontally polarized.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.




NOTE: Figure 1 shall be used by the aircraft manufacturers in deriving installation information.

FIGURE 1. Antenna AT-983/ARN, Outline Of



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

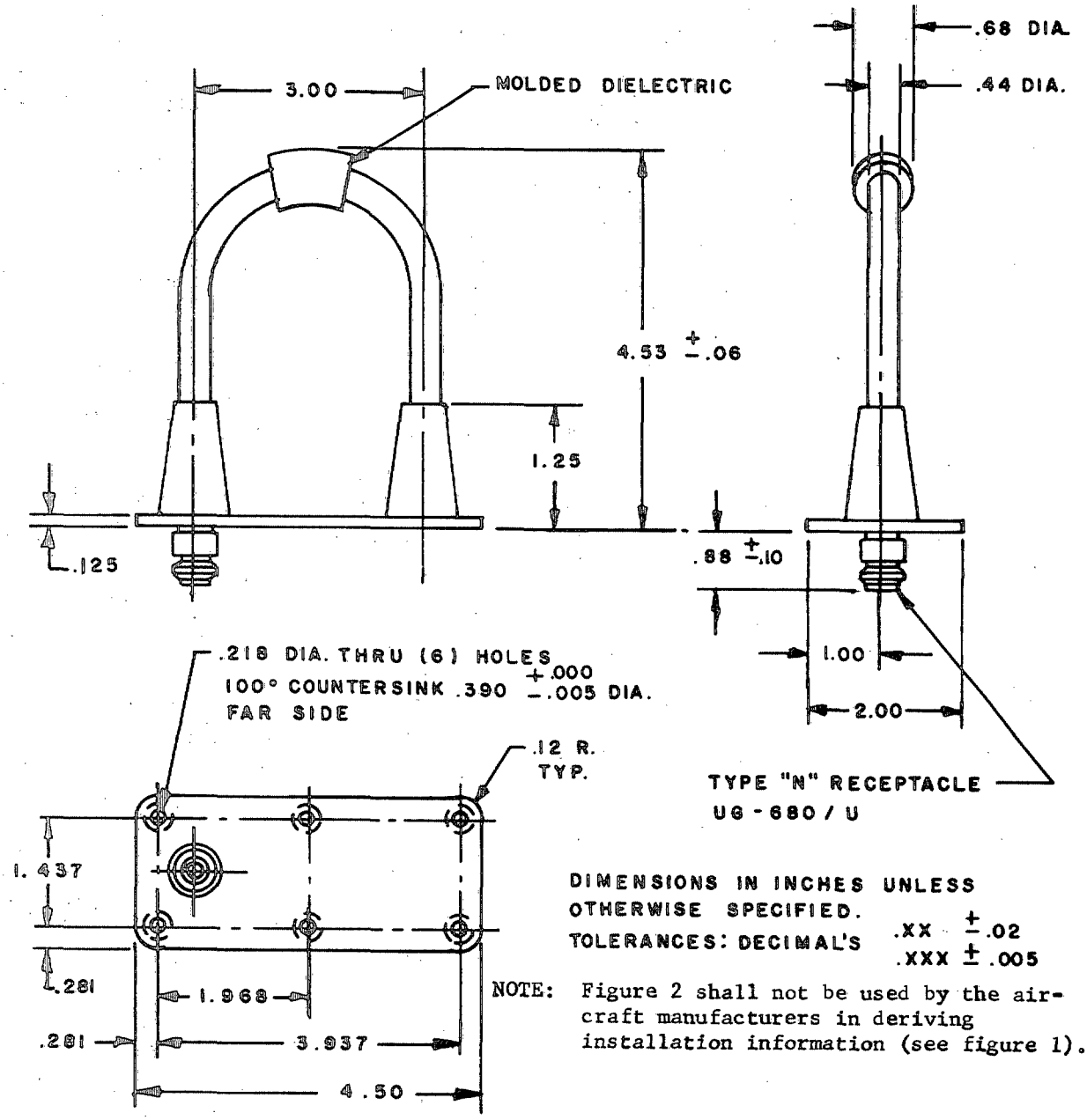



FIGURE 2. Antenna AT-983A/ARN, Outline Of



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

3.4.3 Impedance. The input of the antennas shall match a 52-ohm coaxial transmission line so that the specified maximum of VSWR will not be exceeded over the required frequency range.

3.4.4 Antenna pattern characteristics. When the antennas are tested in accordance with 4.6.2, a pattern shall result with maximum reception within ±10° of 0° where 0° is in line with and directly ahead of the antenna. This pattern shall be obtained for both a horizontal and vertical cut without nulls that exceed 1 dB.

3.5 Government-loaned equipment. The following items will be loaned to the contractor upon his request:

| Item No. | Stock No. | Item Description | Quantity |
|---|---|---|---|
| 1 | 5985-589-2475 | Antenna, Glide Slope AT-983/ARN | 1 |
| 2 | 5826-846-9028CX | Antenna AT-983A/ARN | 1 |

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The quality assurance provisions of MIL-E-5400 and the requirements specified herein shall be applicable to this specification. When the two documents conflict, this specification shall govern.

4.2.1 Classification of inspections. The examining and testing of the AT-983/ARN and AT-983A/ARN antennas shall be classified as follows:

a. Preproduction inspections

b. Quality conformance inspections.

4.3 Test conditions. Unless otherwise specified, the equipment shall be subjected to the tests under the following standard test conditions:

a. Temperature: Room ambient (25°C ±10°C)

b. Atmospheric pressure: 28 to 32 Hg



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

c. Vibration: None

d. Humidity: Room ambient, up to 95 percent relative humidity.

4.4 Preproduction inspections

4.4.1 Preproduction samples. The preproduction samples shall consist of models representative of the production equipment. They shall be tested under the test conditions specified herein and at the point designated by the contract.

4.4.1.1 Testing instructions. The supplier shall conduct tests on at least one preproduction sample to determine that the equipment design meets the requirements specified herein. The supplier's tests shall be conducted in accordance with the test requirements specified under "Test methods" herein.

4.4.2 Preproduction tests. Unless otherwise specified by the procuring activity, one preproduction sample shall be subjected to all the tests specified under "Test methods" herein.

4.4.3 Preproduction test report. A preproduction test report shall be prepared in accordance with MIL-STD-831.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of individual tests.

4.5.1 Individual tests. Each AT-983/ARN and AT-983A/ARN antenna shall be subjected to the following tests:

| Test | Applicable paragraph |
|---|---|
| Examination of product | 4.6.1 |
| Immersion | 4.6.3.4 |

4.6 Test methods

4.6.1 Examination of product. The AT-983/ARN and AT-983A/ARN antennas shall be inspected to verify that the materials, design and construction, necessary mechanical measurements, marking, and workmanship comply with the requirements of this specification, using whatever tools, gauges, scales, and devices that are necessary.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

4.6.2 Antenna pattern test. With the antenna mounted in the test fixture as shown on figure 3, under standard test conditions, horizontal cuts of the antenna pattern shall be plotted over the range of $\pm 90^\circ$ of the dead-ahead position. These plots shall be obtained with the antenna positioned at zero-degree pitch, 20-degree pitch up, and 20-degree pitch down. A single vertical cut of the antenna pattern $\pm 90^\circ$ about the dead-ahead position shall also be plotted. The test antenna ground plane shall be 4 by 4 feet and the minimum separation between the transmitting and the test antenna shall be 18 feet. The objective of this test is to show that the antenna pattern maximum lies within $\pm 10^\circ$ of the dead-ahead position and that there are no nulls that exceed 1 dB relative to adjacent maximums.

4.6.3 Environmental tests. Environmental testing shall be in accordance with the indicated methods of MIL-STD-810 and the requirements specified herein. When the two documents conflict, the requirements of this specification shall govern.

a. The environmental tests may be performed in any sequence except that the immersion test shall be performed last.

b. Only VSWR measurements at frequencies of 328.6, 332.0, and 335.4 MHz shall be required. Specific environment required during VSWR measurements shall be in accordance with the applicable test procedure. Antenna pattern measurements shall be made at 328.6 and 335.4 MHz under the test conditions specified in 4.3.

4.6.3.1 Temperature shock test, method 503, procedure I. Upon conclusion of this test, the antenna shall be positioned in the test fixture as shown on figure 4 and the VSWR shall be measured and recorded at the frequencies specified in 4.6.3b.

4.6.3.2 Humidity test, method 507, procedure I. With the antenna in position on the test fixture as shown on figure 4, the VSWR shall be measured at the frequencies specified in 4.6.3b. These tests shall begin within 10 minutes after removal of the antenna from the humidity chamber.

4.6.3.3 Salt fog test, method 509, procedure I. The output connector shall be plugged to simulate the installed condition. Within one hour of conclusion of this test, the antenna shall be placed on the test fixture in accordance with figure 4 and the VSWR shall be measured at the frequencies specified in 4.6.3b. The antenna shall be examined visually and a record made of any abnormal deterioration. Salt deposits may be removed prior to testing.

4.6.3.4 Leakage (immersion) test, method 512, procedure II. The antenna connector shall not be sealed. Any evidence of leaking shall be cause for rejection. If no leaks are evident, the antenna shall be positioned in the test fixture as shown on figure 4 and the VSWR measurements shall be made at the frequencies specified in 4.6.3b.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

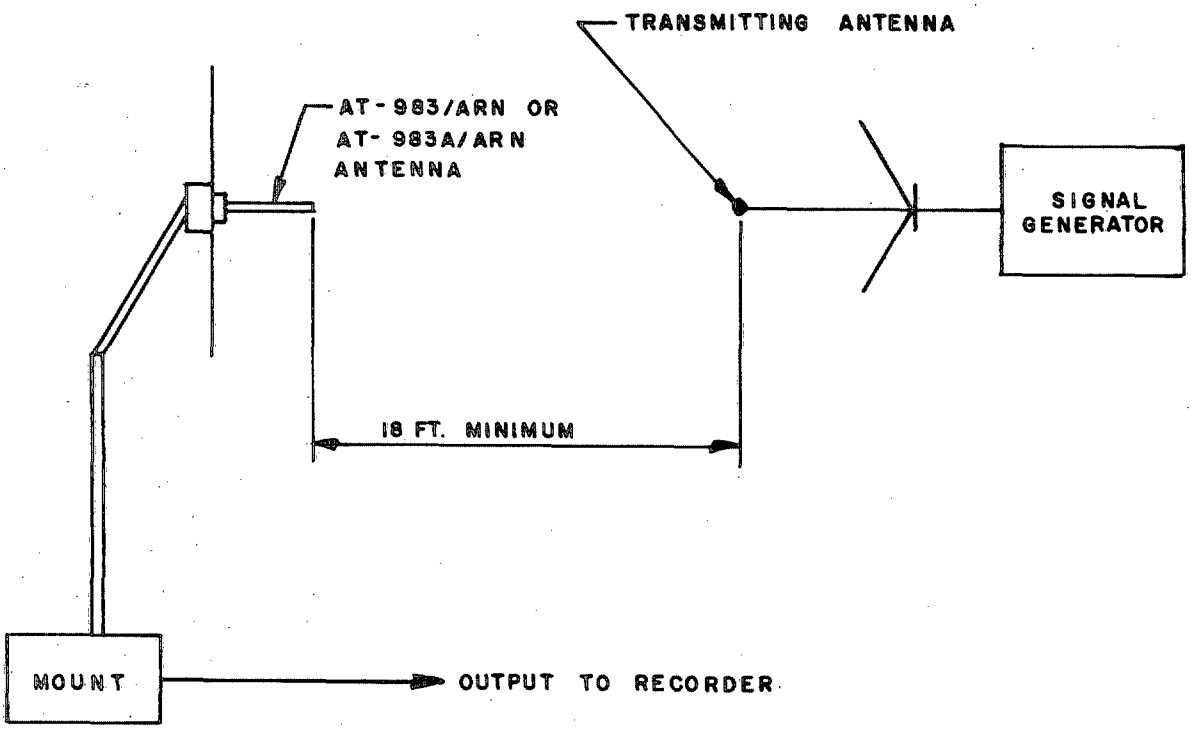



FIGURE 3. Radiation Test Fixture



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

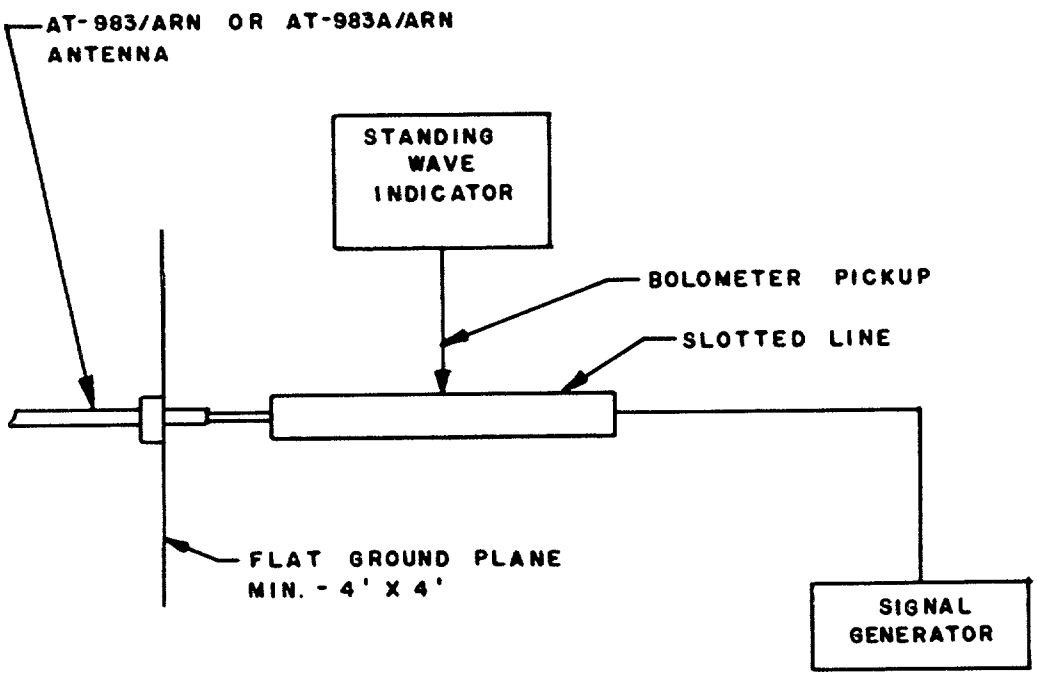



FIGURE 4. VSWR Test Fixture



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

4.6.3.5 Vibration test, method 514.1, equipment category (a), table 514.1-1, procedure I, curves M and AS. The antenna need not be operating during the vibration test. Upon completion of the vibration test, the antenna shall be placed in position on the test fixture as shown on figure 4 and the VSWR shall be measured at the frequencies specified in 4.6.3b.

4.6.3.6 Shock test, method 516.1, procedure I, figure 516.1-1. Upon completion of the shock test, the antenna shall be positioned on the test fixture as shown on figure 4 and VSWR measurements shall be made at the frequencies specified in 4.6.3b.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 The Antenna AT-983/ARN and Antenna AT-983A/ARN shall be prepared for delivery in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use. The AT-983/ARN and AT-983A/ARN antennas are intended to be used to feed ILS glide slope signals to a glide slope receiver.

6.2 Ordering data. Procurement documents should specify the requirements herein.

6.2.1 Procurement requirements

a. Title, number, and date of this specification

b. Preproduction samples and tests

(1) Number of preproduction samples

(2) Point of inspection

(3) Requirement for concurrent delivery of each sample and its test data.

The procuring activity should be given at least 10 days prior notice when the preproduction tests are to be conducted so that they may be supervised or witnessed, if desired, by a Government representative.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

c. Government-loaned equipment (see 3.5).

d. Selection of applicable levels of preservation, packaging, packing, and marking.

6.2.2 Contract data requirements. Data specified in 4.4.3 will be listed directly on a DD Form 1423 incorporated into the contract.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Project No. 5826-F106

☆ US GOVERNMENT PRINTING OFFICE-1972-714-539/1945



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

C16061

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:02Z
Check the source to verify that this is the current version before use.